

VARUN BELAGALI

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Education

Stony Brook University, NY, USA

Aug 2022 – Present

Doctor of Philosophy, Computer Science

Advised by Prof. Dimitris Samaras in Computer Vision Lab

Research Interest: **Visual representation learning with self-supervision**

Coursework: Machine Learning, Computer Vision, Robotics, Distributed Systems, Database Systems (GPA 4/4)

R V College of Engineering, India

Aug 2016 – Aug 2020

Bachelor of Engineering, Computer Science and Engineering

Coursework: Operating Systems, Analysis of Algorithms, Neural Networks, Data Structures, Compilers (GPA 9.22/10)

Research Papers (selected)

1. **TICON: A Slide-Level Tile Contextualizer for Histopathology Representation Learning** [\[Link\]](#) arXiv 2025
Varun Belagali*, S. Kapse*, P. Marza, S. Das,..., P. Prasanna, S. Christodoulidis, M. Vakalopoulou, D. Samaras.
2. **CDG-MAE: Learning Correspondences from Diffusion Generated Views** [\[Link\]](#) arXiv 2025
Varun Belagali, P. Marza, S. Yellapragada, Z. Li,..., J. Saltz, S. Christodoulidis, M. Vakalopoulou, D. Samaras.
3. **Gen-SIS: Generative Self-augmentation Improves Self-supervised Learning** [\[Link\]](#) arXiv 2024
Varun Belagali*, S. Yellapragada*, A. Graikos, S. Kapse, Z. Li, T. Nandi, R. Madduri, P. Prasanna, J. Saltz, D. Samaras.
4. **HyperMAE: Modulating Implicit Neural Representations for MAE Training** [\[Link\]](#) 2023
Varun Belagali, L. Zhou, X. Li., D. Samaras. **NeurIPS Workshop - Self-supervised Learning**
5. **Two step convolutional neural network for automatic glottis segmentation in stroboscopic videos** [\[Link\]](#) 2020
Varun Belagali, A. Rao, P. Gopikishore, R. Krishnamurthy, P. Ghosh. **Biomedical Optics Express**

Experience

Graduate Researcher | Computer Vision Lab, Stony Brook University, NY, USA

Sep 2022 – Present

- TICON: Omni-feature masked modeling for pretraining slide encoder. Multi-teacher distillation in embedding space.
- Self-supervised learning with synthetic data. Gen-SIS: view-invariant SSL with Dino, I-BOT, SimCLR.
CDG-MAE: cross-view SSL with Masked Autoencoders.
- HyperMAE: Implicit neural representation based decoder architecture to efficiently train Masked Autoencoders.

Research Intern | Dolby Laboratories, Sunnyvale, USA

June 2025 – Aug 2025

Post-training Video-JEPA models for egocentric action anticipation.

Research Associate | Indian Institute of Science, Bangalore, India

Oct 2021 – July 2022

Deep Learning for glottis segmentation in stroboscopic videos. Advised by Prof. Prasanta Kumar Ghosh.

Software Engineer | Citrix, Bangalore, India

July 2020 – Sep 2021

Cloud Engineering: Development of traffic manager software to handle regional outages. Lead of cloud cost optimization.

Technical Skills

Technologies: Computer Vision, Deep Learning, Self-Supervised learning, Large-scale Distributed Pre-training.

Languages & Frameworks: Python, PyTorch, Latex

Cloud Engineering: Azure, Jenkins, Splunk, New Relic

Research Talks

Self-supervised Learning with Synthetic Data. **GenBio AI**, Palo Alto, USA (Sep 2025)

Learning Correspondences from Diffusion Generated Views. **DATAIA Talk**, CentraleSupélec, Paris, France (May 2025)

Generative Self-augmentation Improves Self-supervised Learning. **MICS Lab**, CentraleSupélec, Paris, France (March 2025);
IMAGINE Lab, École des Ponts, Paris, France (April 2025)

Fellowships

SBU CS Chairman Fellowship (2024 - 2026), DATAIA Fellowship (2025)